



Social isolation affects brains

Neuroimaging reveals social isolation actually change the brain structure of a person.
<https://digit.in/jul22-14>



TCS' chip-based e-passports

TCS could roll out chip based e-passports by the end of 2022.
<https://digit.in/jul22-22>

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EARTH

Playa Grande beach



tar was found on the surface of the rock. At the Famara beach, the local people noticed incidents where tar was attached to rock surfaces. These formations were mainly observed on the rocky coast of Famara village, which is on one side of the main beach. During the examination the coast of Tenerife beach within the confines of the same project, similar formations were found along the coast of Malpaís de Güímar. The formations on these beaches were a result of tar attached to the surface of the rocks over time. Materials the sea brought to the shore helped these formations amass.

A THREAT BEYOND THE SHORES

Apart from choking life forms, the plastitar has other very concerning effects on the environment. When the sun is out, it will more than likely increase the temperature of the ground, putting the lives of other species around the affected rock at risk. A study conducted on Henderson Island showed how plastic contributed to the increasing the temperature of the beach grounds. Sea turtles, for one, are directly affected. The

gender of a sea turtle is dependent on the temperature of the surface the egg is laid on; at higher temperatures, the offspring turn out to be female. A change in the species' gender ratio could prove to be catastrophic.

Another potential risk posed is the release of harmful chemicals from plastitar. This could harm the marine food chain, which is not a good sign. It is still uncertain which species will be harmed by these toxic chemicals but they will cause a blockage in environmental development.

Plastitar adds another layer of complication to the plastic pollution problem. For a long time, environmentalists were preoccupied with visible issues, such as floating bottles and bags. It wasn't until the early 2000s that scientists began a serious investigation into microplastics, discovering that almost the entire planet is contaminated. These particles are in the atmosphere, making their way to the highest mountains. They contribute to climate change, though it's unclear whether they will ultimately help heat or cool the planet. We know that people consume large amounts of microplastics, and babies consume

even more in their formula, but scientists are only now beginning to look into the impact on human health.

PLASTITAR IS NOT THE ONLY ONE

Even more recently, scientists have discovered "new plastic formations," of which plastitar is only the most recent. For example, when plastic burns in beach campfires, it forms a gnarly matrix of polymer mixed with sand and other debris. High outdoor temperatures can cause the synthetic material to blend into the rocks, forming this 'Plasticrust'. Scientists are also looking into anthropoquina – sedimentary rock made of plastic and other man-made materials.

NO END IN SIGHT

Plastics have been in circulation since the 1900s and our annual production across the planet has been on a gradual incline since the 1950s. In the last 70 years, as much as 10 billion tonnes of synthetic polymers or plastics have been produced globally. Most plastic waste is dumped into landfills, incinerated or gets recycled. An estimated 12-18 per cent of global plastic waste gets deposited into the environment due to mismanagement and common littering.

The production of plastic waste has increased twofold in the last two decades with no end in sight. We have also seen a 4x increase in plastic waste as compared to what the planet generated 30 years ago. Around 3 million tonnes of microplastic waste is generated every year. This excludes about 5 million tonnes of macroplastics that eventually break down into microplastics. Considering only 9 per cent of plastic waste is being properly recycled, people need to take a much closer look at plastic waste management.

Mismanaged plastics have slowly become part of the earth's state of being, much like nitrogen, carbon and phosphorus are part of the planet's core biogeochemical cycles – and a global plastic cycle doesn't bode well for us.